Local goodness-of-Fit measures for neural decoding

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Abstract

State-space models are widely used in many fields to infer latent states from observations, but there are still some issues and challenges related to them. First, because latent variables are inherently unobservable, their decoding results cannot be directly validated against observed data. Second, it is often challenging to disentangle prediction errors stemming from model misspecification. Third, such errors can accumulate and propagate over time, amplifying uncertainty in long-term predictions. Moreover, many state-space models are purposefully simplified, accepting some degree of inaccuracy in exchange for greater computational efficiency. To address this challenge, we developed three classes of metrics designed to identify localized periods where model misspecification leads to inaccurate estimates. Each metric assesses the consistency between the information contained in individual observations and the information extracted by the state-space model from a local window of data. These metrics fall into three

categories: 1) a set of divergence measures, specifically F-divergences, which compare the distributions of the latent state process directly; 2) measures of overlap between estimated credible regions obtained from the state distributions; and 3) predictive checks for individual observations, based on samples from the estimated state process. We applied these metrics to the problem of estimating nonlocal spatial representations in both simulated and real hippocampal spiking data, demonstrating the strengths and limitations of our approach.

1 Introduction

Many brain states, such as those related to memory, emotion, and attention, are not directly observable. They are internal states that are not solely determined by the external world and can change over time. This makes them challenging to understand. However, these brain states are critical for understanding how the brain supports complex behavior, because they go beyond simple stimulus and response relationships. A major challenge for systems neuroscience is to relate neural activity (spikes, spike waveforms, calcium traces, etc.) to these latent dynamic brain states.

State space models are a powerful way to understand how neural activity relates to latent dynamic brain states. They have been applied to diverse neural systems such as gustatory activity [Jones et al., 2007], hippocampal replay dynamics [Denovellis et al., 2021], and hypothalamic activity dynamics [Vinograd et al., 2024]. The core assumption of these state space models is that complex, high dimensional neural activity can be related to low dimensional, latent states. State space models aggregate information over time about the latent states, weighing both the information from the data at the current time (normalized likelihood distribution) with the accumulated information used to predict the latent state (prediction distribution). This powerful combination enables robust estimation of the latent state, effectively balancing current information with the accumulated history of the system and prior knowledge.

Despite the wide use of state space models in neuroscience, methods to validate state space models remain underdeveloped and underutilized. One reason is that latent states are inherently difficult to evaluate [Newman et al., 2014]. Without ground truth, model validity can only be inferred indirectly—from relative likelihoods under multiple models, from the quality of reconstructed signals, or from performance on downstream tasks. Another challenge is that neural data are dynamic, so model fit may vary substantially across time. Compounding this problem is that common methods of validating models, such as cross-validation and "shuffle" permutation tests, are holistic evaluations of the model – they tell neuroscientists whether or not the model generalizes well or is different from a simplified model with less utility – but they fail to indicate when the model fits poorly or which specific components are responsible. Consequently, these methods can reject models that are intentionally simplified yet scientifically meaningful and offer little guidance for targeted improvement of the model. Many state-space models in neuroscience are intentionally simplified—not because the brain is simple, but because tractable computation and clear interpretation often matter most. For example, some state-space models smooth emotional movement into a low-dimensional trajectory [Deng et al., 2015]; others compress field-linked structure into a stationary, locally anchored map [Hsin et al., 2022]. Some frameworks simplify both at once, jointly compressing affective trajectories and local field potentials into compact latent states for computation and inference [Tao et al., 2018, Zeng and Eden, 2025]. Under standard goodness-of-fit criteria, such simplifications would likely fail.

State space models already contain rich diagnostic information in the form of distributions used in their estimation. One potential solution for more targeted local diagnostic information of state space models would be to utilize the information in the normalized likelihood, which reflects information from only the current neural observations, and prediction distribution, which aggregates information from all past observations based on the state space model structure. When these two distributions agree, the model's prior expectations and data-driven evidence are consistent. When they diverge, the mismatch can reveal where and when and potentially how the model fails to capture the structure of the data. Examining these discrepancies could therefore provide a direct, interpretable measure of model–data agreement—analogous to a local, time-resolved goodness-of-fit test.

To address these gaps, we develop and validate a suite of three complementary goodness-of-fit methods – KL divergence, highest posterior density (HPD) overlap region, predictive checks – designed to provide actionable feedback to neuroscientists. Specifically, each of these methods utilize the mismatch between prediction and likelihood to: (1) identify time periods of poor model–data agreement; (2) distinguish errors arising from the state versus observation components; and (3) provide intuitive visualizations accessible to a broad audience. We demonstrate the utility of these techniques on simulated and real hippocampal data, offering a path for neuroscientists to move from binary model rejection toward an iterative, diagnostic approach to building interpretable theories of neural dynamics.

2 Methods

2.1 State-Space Models

The state-space paradigm is particularly well suited for neural data analysis since many neuroscience experiments focus on internal brain processes that are not directly observable. For example, hippocampal replay of spatial trajectories has been hypothesized to be related to mental exploration of space, but the specific spatial trajectories being replayed are not directly observable by experimentalists [Tao et al., 2018]. In such cases, state-space models are used to estimate simultaneously the unobserved internal representations and their relation to neural activity patterns.

For a discrete set of time points, $t_0, \ldots, t_K$, we define an unobserved state process x_k, which influences a neural observation process y_k at each time t_k. We define a state space model by combining a state transition distribution $p(x_k|x_{k-1})$, which defines how the state evolves in time, with an observation likelihood $p(y_k|x_k)$, which defines the neural representation of each state. To simplify the problem, we assume that the state transition depends only on the most recent past value of the state and that the observation likelihood depends only on the current state value. Figure 1 (a) shows a schematic of this state-space model structure.

This state-space model structure allows for iterative estimation of the state process at each time step based on all the neural data up to that time, as shown in Fig. 1 (b). If the distribution of the state at time t_{k-1} has been computed, we can convolve it with the state transition distribution to obtain a one-step prediction distribution, $p(x_k|y_{1:k-1}) = \int_{x_{k-1}} p(x_k|x_{k-1}) \cdot p(x_{k-1}|y_{1:k-1}) dx_{k-1}$. This predictive distribution combines the information from all previous observations to compute the distribution of the state at the next time step. We update this predictive distribution with the information from the most recent observation by multiplying by the observation likelihood to compute the posterior distribution of the state given all of the data up to the current timestep, $p(x_k|y_{1:k}) \propto p(y_k|x_k) \cdot p(x_k|y_{1:k-1})$.

Common approaches for assessing the goodness-of-fit of state-space models include examining the cross-validated error measures in the state process when the state is observable [Stone, 1974], cross-validated error measures in the observation process such as the mean-squared prediction error [Hastie et al., 2009], and the distribution of prediction residuals [Shen and Zhuang, 2024]. For many neural modeling problems, state-space models are intentionally simplified and these standard goodness-of-fit measures trivially show lack of fit.

Here we explore a set of local goodness-of-fit measures, each of which compares the prediction distribution to the normalized likelihood of the most recent observation as a function of the state or to that observation itself. Thus, these measures consider the consistency between information accumulated through the state-space model over past spiking with the information from the most recent spiking activity. We do not expect these distributions to align perfectly since they likely contain different amounts of information about the state process. Instead, we assess whether the information these distributions provide about the state and observation processes is consistent in the sense illustrated in Fig. 1 (c). Specifically, we say that the two distributions are inconsistent when their probability mass is concentrated in mostly non-overlapping regions of the state space. This suggests disagreement between the information from past spiking and

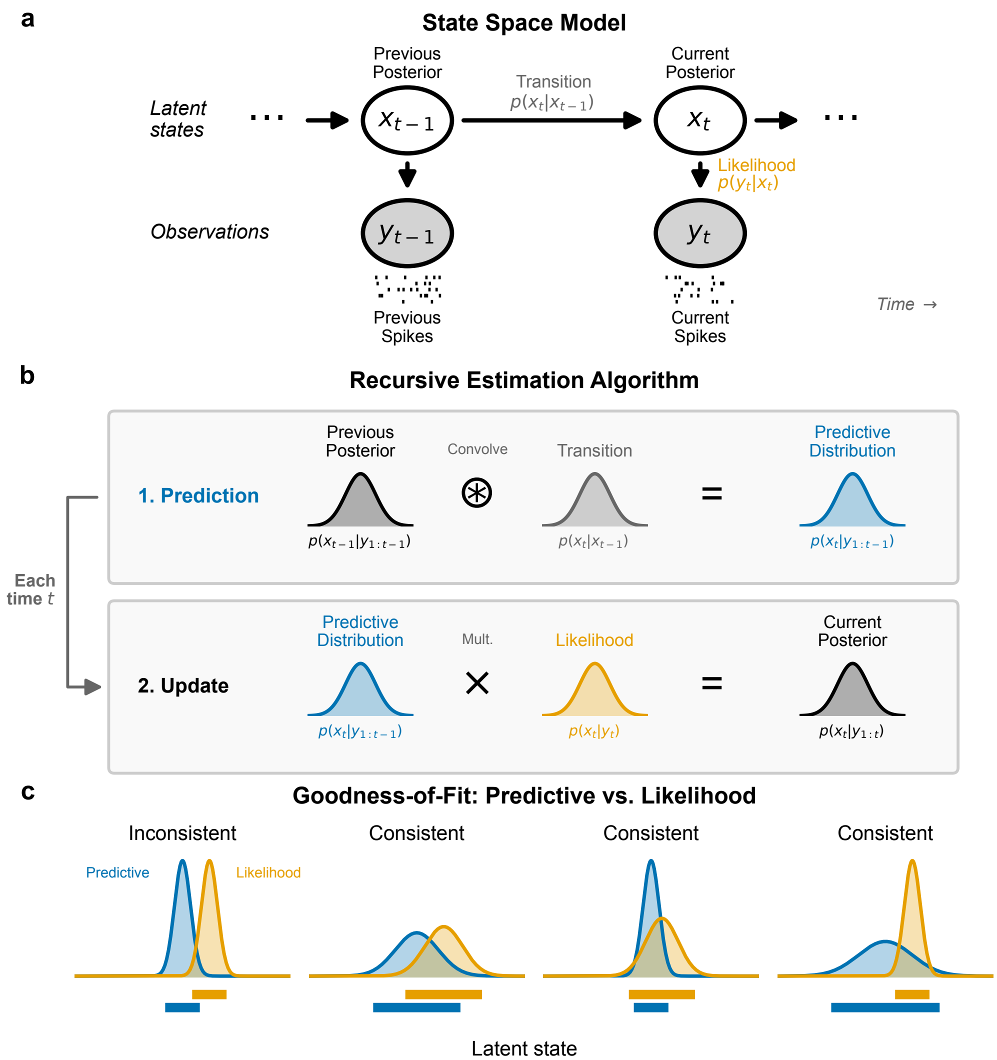

Figure 1: Schematics for state-space model and goodness-of-fit measures. (a) Graphical model depiction of state-space model includes state transition model, $p(x_t|x_{t-1})$, and observation likelihood $p(y_t|x_t)$. (b) These model components are sufficient to compute an iterative expression for the posterior filter distribution at time t_k, $p(x_k|y_{1:k})$ from the distribution at the previous time step. This involves first convolving the previous posterior with the state transition distribution to compute the one-step predictive distribution, $p(x_k|y_{1:k-1})$, and then updating the predictive distribution with information from the most recent spiking by multiplying by the observation likelihood. (c) For each new spike observation, we compute the local goodness-of-fit by assessing the consistency of information between the predictive distribution and likelihood. These are considered inconsistent when they assign high probability to mostly disjoint subsets of the state-space. When the regions with high probability overlap, the distributions are considered consistent, even if the distributions differ in their central tendency or certainty.

the information from the most recent spiking. Distributions that largely align, or that are shifted from each other but are supported over largely overlapping regions of state space are considered consistent. Additionally, if one distribution covers a broad region of the state-space and the other covers a much narrower subset of that region, the distributions are considered to be consistent. We expect this situation to arise often since the prediction distribution contains information accumulated from many past spikes.

2.2 Local Goodness-Of-Fit Measures

In this section, we introduce three classes of metrics designed to characterize the local goodness-of-fit of the state-space model.

2.2.1 KL Divergence

One of the most well-known measures of discrepancy between probability distributions is the Kullback-Leibler (KL) divergence [Kullback and Leibler, 1951], defined in this case as the expected excess self-information (log of the density function) when using the likelihood to predict the state in place of the predictive distribution,

$$D_{\mathrm{KL}}(P\|Q) = \int P(x_k) \log\left(\frac{P(x_k)}{Q(x_k)}\right) dx_k, \tag{1}$$

where $P(x_k) = p(x_k|y_{1:k-1})$ is the predictive distribution and $Q(x_k) = \frac{p(y_k|x_k)}{\int p(y_k|x_k)dx_k}$ is the normalized likelihood in x_k given the observed spiking y_k.

Figure 2 (a) illustrates the computation of the KL divergence. The top panel shows the predictive distribution and normalized observation likelihood as functions of x_k. The middle panel shows the difference in self-information between these distributions, $\log\left(\frac{P(x_k)}{Q(x_k)}\right)$. The bottom panel is this difference in self information weighted by $P(x_k)$, which when integrated gives the value of the KL divergence.

The KL divergence is an asymmetric measure, meaning that $D_{KL}(P||Q) \neq D_{KL}(Q||P)$.

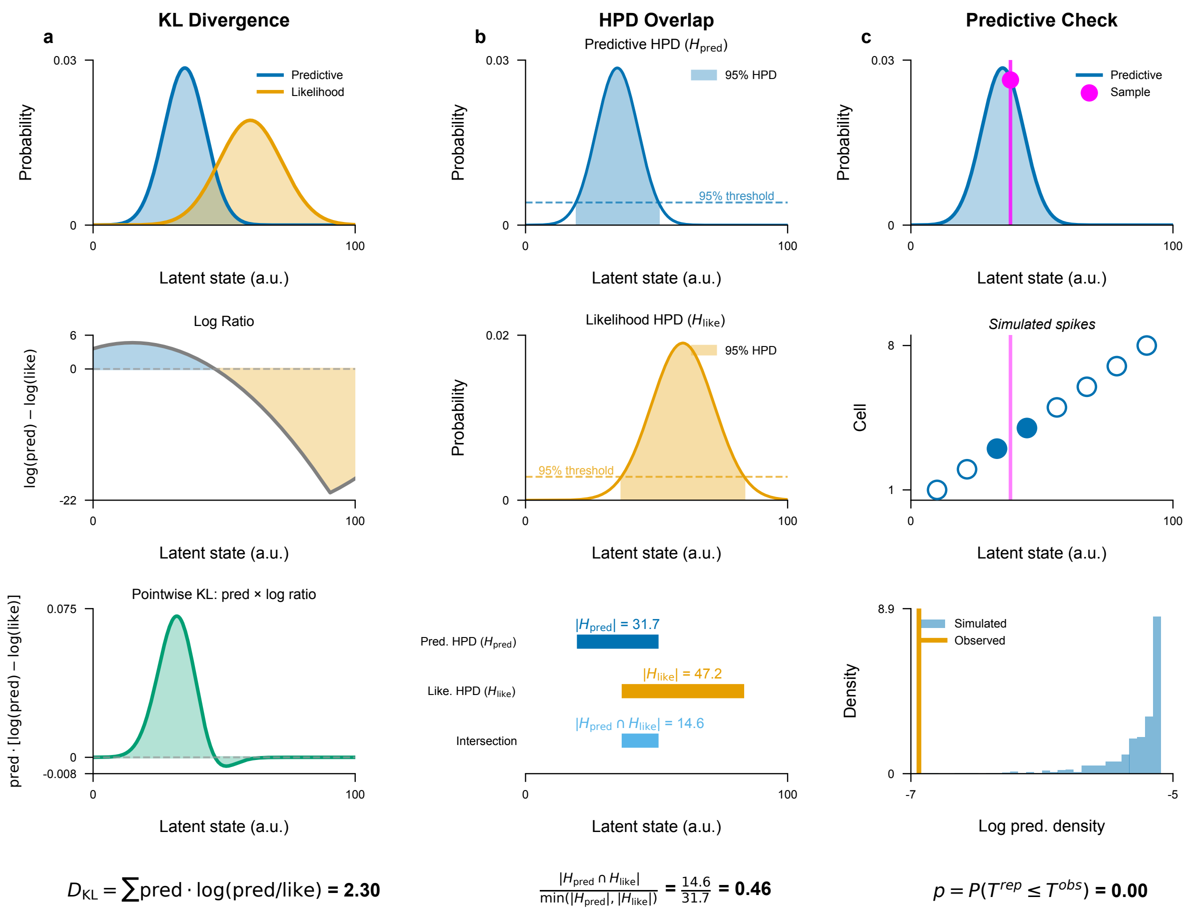

Figure 2: Visualizations of the computation of each local goodness-of-fit metric. (a) The KL divergence compares the predictive distribution (blue) and the observation likelihood (orange) (top panel) by computing the difference in their log density (middle panel) and computing its expectation with respect to the predictive distribution (bottom panel). (b) The HPD overlap is defined by computing the HPD for the predictive distribution (top) and for the observation likelihood (middle) and then computing the size of their intersection relative to the size of the smaller of these two HPD regions (bottom). (c) Predictive checks are computed by sampling values of the state from the predictive distribution multiple times (top), then sampling from the conditional distribution of neurons that could have generated the observed spike given each sampled state value (middle), and then computing the probability that the log of the predictive distribution of the observed spike is greater than the log of empirical predictive distribution of the sampled spikes (bottom).

We could also compute the KL divergence from the likelihood to the predictive distribution $D_{KL}(Q||P)$ or the mean between these two divergences $\frac{1}{2}(D_{KL}(P||Q) + D_{KL}(Q||P))$. More broadly, the KL divergence is part of a larger family of divergence metrics between distributions, known as F-divergences. Other members include the total variation distance [Bhattacharyya et al., 2023] and the Pearson χ^2 divergence [Crack, 2018]. It is simple to extend the KL divergence metric between the predictive distribution and the observation likelihood to any of these other divergence metrics to identify differences in additional features of the distributions.

2.2.2 Overlap Between HPD Regions

Another way to assess the consistency between the predictive distribution and the observation likelihood is to examine the extent to which their highest posterior density (HPD) regions overlap. A substantial intersection suggests that the two distributions convey consistent information, reinforcing the reliability of the inferred latent state. In contrast, minimal or no overlap signals a significant discrepancy, hinting at potential inconsistencies in the local fit of the model. This approach provides an interpretable and intuitive metric of the degree of consistency between these distributions.

We define the overlap between the HPD for the predictive distribution and the likelihood as

$$\text{HPD Overlap}\,(P, Q) = \frac{\left|\cap\left(HPD_P, HPD_Q\right)\right|}{\min\left\{\left|HPD_P\right|, \left|HPD_Q\right|\right\}} \tag{2}$$

where HPD_P and HPD_Q are the highest posterior density regions of the predictive density and the normalized observation likelihood respectively and $| * |$ is the volume of the region. The numerator is the size of the overlap between these regions and the denominator is the size of the smaller of the two HPD regions. This normalization ensures that the resulting value is in the range $[0, 1]$, with 0 indicating no overlap, and 1 indicating that one of the HPDs is entirely contained within the other. This means that if, for example, the likelihood produces an HPD that is wider and therefore less informative than the predictive density but that completely contains the HPD of the predictive density then the HPD ratio will be 1, suggesting that the two distributions are statistically consistent. Conversely, if the ratio is close to 0, it indicates little or no overlap, suggesting significant differences between range of values for the state that would be inferred from these two distributions.

Figure 2 (b) illustrates the computation of the HPD overlap. The HPD for the predictive distribution (top) and the normalized likelihood (middle) are computed, and the size of their intersection is normalized by the size of the smaller of the two HPDs (bottom).

2.2.3 Predictive Checks

The previous metrics compared two distributions of the state: one based on the accumulated past information and the other based on the isolated information from the likelihood of the most recent spiking observation. Another approach is predictive checking, which compares the observation itself to a predictive distribution of that observation estimated from the model. García and Castellanos [2007] proposed Bayesian checking of the second levels of hierarchical models with partial posterior predictive checking. Moran et al. [2019] used population predictive checking to evaluate model's goodness-of-fit. Guttman [1967] used future observations to evaluate model's goodness-of-fit. Predictive checks have the advantage that they allow for the computation of p-values for individual observations, which are easy to interpret [Meng, 1994, Gelman et al., 1996].

Predictive checks for spiking data can be challenging to implement, since in sufficiently small intervals, any spikes are unlikely compared to no spiking at all. One approach to deal with this might be to consider patterns of spikes over longer fixed intervals over which multiple spikes would be expected. Here, we take a simpler approach of focusing only on spike times and measuring the likelihood that an observed spike comes from a particular neuron (for spike-sorted models) or has a particular set of waveform features (for clusterless models).

In particular, let $y_{k,j}$ be the identity of the neuron (spike-sorted model) or a spike waveform feature (clusterless model) for the j^{th} spike occurring in the k^{th} time bin. We define the predictive distribution of the observation based on the state model as

$$f_{\text{predictive}}\left(y_{k,j}\right) = \int p\left(y_{k,j} \mid x_k\right) p\left(x_k | y_{1:k-1}\right) dx_k \tag{3}$$

to be the predictive distribution of $y_{k,j}$, where $p\left(y_{k,j} \mid x_k\right)$ is the observation process and $p\left(x_k | y_{1:k-1}\right)$ is the one-step prediction distribution of the current state given all previous spikes. We can sample new observations, $\widetilde{y}_{k,j}$ from this predictive distribution, and compute the probability that the observed spike is as likely as the sampled spike.

$$D_{k,j} = Pr\left(\log\left(f_{\text{predictive}}\left(y_{k,j}\right)\right) \geq \log\left(f_{\text{predictive}}\left(\widetilde{y}_{k,j}\right)\right)\right). \tag{4}$$

The log terms in Equation (4) are unnecessary, but suggest the method to estimate $D_{k,j}$. We sample x_k multiple times from the one-step prediction distribution, $p\left(x_k | y_{1:k-1}\right)$ and use these to generate samples of the spike observations, $\widetilde{y}_{k,j}$ from the predictive distribution. We then compute the fraction of these samples whose log likelihood is smaller than or equal to that of the observed spike.

Figure 2 (c) illustrates the computation of a predictive p-value for spike sorted data. Many values of the state are sampled from the predictive distribution (top), each of which is used to sample a neuron that could have generated the observed spike (middle). The log likelihood of the observed spike is then compared to an empirical cumulative distribution of the log likelihoods of the sampled spikes (bottom) leading to a goodness-of-fit metric that can be interpreted as a p-value for that individual spike. High p-values suggest that the observed spike is consistent with the predictive distribution of spikes and a p-value near zero suggests that the observed spike is unexpected based on the predictive distribution computed from the past spiking and the model.

3 Simulation Study

To demonstrate the performance of our local goodness-of-fit metrics in real-world applications, we designed a simulation study, generated neural signal data, constructed state-space model, and applied our metrics to evaluate the model's local goodness-of-fit. We established a framework to analyze spatial mapping in the hippocampus of a rat. Specifically, we simulated data that approximates the rat's movement and the corresponding hippocampal spike activity. Our simulation will encompasses both local and non-local states, which, while resembling real-world rat behavior and neural signals to some extent, may also deviate from them. Our goal is to decode the latent variables using an intentionally simplified model applied to data generated by a more complex, true generative process. We then utilize local goodness-of-fit metrics to assess the model's alignment with observed data, identifying periods of potential misspecification and evaluating the model's reliability over time.

3.1 Generative Model

3.2 Simulation Results

4 Real Hippocampal Data Analysis

5 Discussion & Conclusion

State space modeling has become a powerful tool for describing the latent dynamics of neural systems. However, validating these models is inherently difficult because latent states are unobservable and many models are purposefully simplified. To address this, we developed multiple local goodness-of-fit metrics that compare the instantaneous information received from each spike to the aggregated information computed based on recent spiking activity and the state-space model assumptions. Rather than dismissing a model in its entirety, these metrics are able to assess the consistency between immediate observations and the information accumulated through the state-space structure, providing a more granular assessment of decoding reliability.

We defined three classes of metrics: F-divergence measures (e.g., KL divergence), overlap ratios between credible regions (HPD overlap), and predictive checks. While KL divergence is computationally simple, it is sensitive to distribution tails and lacks a clear upper bound for assessing similarity. In contrast, the HPD overlap offers a more robust geometric measure of similarity, consistently maintaining lower error rates across various conditions. Predictive checks provide a direct statistical assessment through observation-level p-values, but are more computationally intensive. While each metric has particular advantages, when used in combination, they provide a more complete perspective on decoding reliability.

We illustrated the application of these metrics to simulated and real spike sorted data from rat hippocampus. These same metrics naturally extend to other neural data types when incorporated into state space models. In particular, recent work has shown that clusterless neural spiking models can substantially reduce bias and variance in neural decoding and estimation problems when compared to spike sorted models. The metrics, as defined in Section 2, naturally extend to clusterless models. In that case, a lack of fit for any particular spike suggests that a spike with that waveform was unexpected based on the accumulated information from the state space model up to that point in time.

For simplicity, in this paper, we focused on comparing the instantaneous information in the likelihood of each spike to the aggregated information from the past contained in the one-step prediction density. A natural, powerful extension of these methods is obtained by replacing the one-step prediction density with any posterior density derived from the state-space framework. In particular, replacing the one-step prediction density with a smoothing density (e.g. from a fixed-interval smoother over the observation period) would allow us to leverage both past and future data to increase statistical power in identifying model misspecification.

One potential challenge for applying these methods broadly has to do with the computational burden of computing these metrics as the dimensionality of the state process increases. In all of our examples, we computed the KL divergence measure and the HPD overlap by partitioning the state space into a lattice and computing the likelihoods and prediction densities at each point in that lattice. As the dimension of the state increases, the size of that lattice will grow exponentially. While the predictive checks are the most computationally burdensome in lower state dimensions, the fact that we are sampling from the state-space rather than computing values over a lattice means that in higher dimensions the predictive checks will become more computationally efficient. It also suggests that we may be able to improve the efficiency of the KL and HPD overlap metrics by using sampling-based estimation methods. This will not solve the curse of dimensionality, as the number of samples required for accurate estimates will likely still grow with dimension, but it is likely to allow for efficient estimation in moderate dimensions. Another approach for dealing with this issue may be to compute these consistency metrics in lower-dimensional subspaces of the state space. How to identify subspaces that retain the statistical power of the original space is a topic for future exploration.

Another potential challenge relates to identifying time-scales for which to assess for lack of fit. These methods allow us to focus in on time scales as short as individual spikes, and for our place cell decoding examples individual spikes are sufficiently informative that we are often able to detect local model misfit at this time-scale. For other neural systems, it may be the case that local model errors are more likely to be detected over longer timescales. In this paper we discuss how to extend these goodness-of-fit measures to longer time intervals, but determining what time scales are most appropriate may require relying about prior knowledge about the system, or exploring the values of these metrics across multiple time scales.

We believe that these local goodness-of-fit tools provide an important step toward making inferences from state-space neural models more reliable and interpretable. By pinpointing specific moments when the model assumptions break down, researchers can be more confident in their inferences. Ultimately, these measures empower us to refine our understanding of neural dynamics and deepen our understanding of how the brain encodes and processes information.

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